

APPLICATION WORKING GUIDE

SashaInfinity Campus OS

A complete, module-by-module working guide to the SashaInfinity learning and campus platform: purpose, key features and step-by-step workflows for learners, instructors, campus teams, companies, parents and administrators.

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PART 1

Platform Overview and Access

SashaInfinity Campus OS is a learning management system extended into a shared operating workspace for schools and colleges. It combines an online course platform (video lessons, quizzes, assignments, interactive labs, games and live classes) with the daily operations of an institution: admissions, fee collection, exams, staff management, transport, hostel and guardian communication. Every person signs in once and lands in a workspace tailored to their role.

Role	What this role does	Landing page
Student	Learns: courses, labs, live classes, plans, mastery, internships	/dashboard
Instructor	Teaches: courses, assessments, content studio, live classes, insights	/instructor/dashboard
Institution owner / admin	Runs the campus: admissions, finance, exams, staff, transport, hostel	/institutions
Teacher (campus)	Teaches batches, enters marks, applies for leave	/institutions
Parent / guardian	Monitors children: attendance, fees, results, transport, hostel	/parent
SPOC	Manages college placement cohorts and internships	/spoc/dashboard
Company / Company manager	Runs company portal: internships, work logs, seat pools	/company/dashboard
Admin	Operates the platform: people, content, commerce, communications	/admin/dashboard
Superadmin	Supervises admins and audits impersonation	/superadmin/dashboard

Table 1. Roles, responsibilities and landing pages.

1.1 Signing in and managing your account

The account module covers registration, sign-in, verification and personal settings. Authentication uses short-lived JWT access tokens with refresh tokens; admin and superadmin accounts additionally require a TOTP authenticator code.

Key features

- Email and password sign-in with an optional Remember me setting.
- Email verification for newly registered accounts.
- TOTP two-factor authentication for platform administrator accounts.
- Profile page: name, avatar, password change and instructor public profile.
- Communication preferences: choose which email and notification categories you receive.
- Password reset by email link; Google and LinkedIn sign-in buttons appear when configured.

How to work in this module

1. Open the site and select Workspace (or go to /login).
2. Enter your email and password, then press Sign in. Admin accounts are then asked for the 6-digit authenticator code.
3. Open the avatar menu (top right) to reach Profile and Communication preferences.
4. In Profile, update your details or change your password; use Sign out when finished.

A session that expires returns you to the login page automatically. If Google sign-in shows as not configured, use email and password.

1.2 The role-based workspaces

After sign-in, the sidebar groups destinations by the job each role is doing, so every feature is at most two clicks away. Active groups open automatically, and live badges (grading counts, pending approvals, unread messages) decorate the relevant items.

Key features

- Students: Learning, Progress, Explore, Career and Community groups.
- Instructors: Teaching, Content Studio, Assessment and Support, Learners and Insights groups.
- Admins: Learning and Content, People and Access, Commerce, Internships, Insights and Community groups.
- Campus roles (owner, admin, teacher, student members) share the institution workspace with tabs filtered by role.
- Parents, SPOCs, companies and superadmins get compact role-specific menus.

How to work in this module

1. Sign in and observe your sidebar; the groups you see are your permission map.
2. Click a group to expand it; the compact rail keeps collapsed groups one click away.
3. Watch for count badges - they mark work waiting for you (grading, approvals, messages).
4. Use the search box in the header to find learning content quickly.

PART 2

Public Site and Storefront

2.1 Home page

The home page is the public storefront: it explains the product, surfaces categories, programs, testimonials, instructors and the newsletter, and routes visitors into learning.

Key features

- Hero section with quick links: Explore courses, Try a learning lab, Explore subjects, Find an internship, Read and discover.

- Category tiles for the Tamil taxonomy (Meiporul, Seyappaduporul, Utporul).
- Learning opportunities strip: Learn, Experiment, Read, Build experience.
- Testimonials carousel, instructor highlights, partner logos and newsletter signup.
- Global search box (Search courses) available from every public page.

How to work in this module

1. Start at the home page and use the hero links or the navigation bar.
2. Scroll to Browse by Categories and open a category to see its courses.
3. Use the footer for company information, policies, bundles and membership pages.

2.2 Course catalog and course detail

The catalog lists every published course with filters; the course detail page is the sales and preview surface. Visitors can preview marked lessons without an account, but locked lesson content is stripped from the payload server-side.

Key features

- Filters: category, learning format, level and price; search matches title, description and tags (including similar tag clusters).
- Course cards show rating, lesson count and price, with sale prices displayed when active.
- Course detail: curriculum list with per-lesson lock state, instructor profile, reviews and related bundles.
- Preview lessons open a public reader; is_locked markers show what requires enrollment.
- Wishlist button saves courses for later; cart button starts a purchase.

How to work in this module

1. Open Courses in the navigation and refine with the filter sidebar.
2. Open a course to read the description and curriculum; lessons marked preview can be opened immediately.
3. Add the course to the cart or wishlist, or enroll directly if it is free.
4. Check the reviews and instructor link before buying.

Content is locked server-side: locked lessons never expose video URLs or answers, even in the page source.

2.3 Bundles and memberships

Bundles package several courses into one one-time purchase. Memberships are recurring subscription tiers that grant access to a catalog of courses while the subscription stays active.

Key features

- Bundles list and detail pages with per-course breakdown and one-time pricing.
- Membership tiers with price comparison; plan changes create new Razorpay plans (prices are immutable per plan).

- Coupons apply to single courses; bundles and seat-priced cohorts reject them.
- Edge caching for anonymous visitors keeps these pages fast under load.

How to work in this module

1. Open Bundles or Membership from the footer or course page promos.
2. Compare what each bundle or tier includes.
3. Add to cart and check out once; access is granted automatically after payment verification.

Enrollments created by a membership are marked with a special source and are suspended automatically if the subscription lapses. Paid single-course enrollments are never touched.

2.4 Digital library (eBooks)

The digital library is a separate catalog of eBooks that can be sold or read online. Purchased titles appear under My Library for reading.

Key features

- Library listing with categories and search.
- Book detail pages with cover, description, sample and price.
- Reader opens purchased eBooks; My Library collects everything you own.
- Library payments share the cart and Razorpay pipeline with courses.

How to work in this module

1. Open Library from the navigation.
2. Open a title to preview it, then purchase or add to cart.
3. Read your titles any time under My Library in the student sidebar.

2.5 Learning labs

Learning labs are interactive science and mathematics simulations - physics, chemistry, biology, science, maths and XR - curated from a built-in catalog plus admin-imported packs. They are freely explorable and also embedded in lessons.

Key features

- Lab library grouped by subject with search and curriculum mapping (CBSE / NCERT 2024).
- Concept labs pair a simulation with an investigation worksheet and checkpoints.
- 3D and WebXR labs (solar system, molecules, atom builder) with device support detection.
- Native labs (reaction balancing, identification) validate answers server-side.
- Spatial mode for classrooms projects one lab to a shared display while students observe.

How to work in this module

1. Open Labs from the home page or sidebar.

2. Filter by subject or search; open a lab card.
3. Follow the on-screen instructions; checkpoints record your progress when you are signed in.
4. Return any time - results feed your mastery profile when the lab is part of a course.

2.6 Practice papers and exam papers

The paper generator assembles printable exam papers from question banks. Public users can browse exam papers; priced papers are purchased through the same store.

Key features

- Blueprint-driven generation: choose subject, chapters, difficulty and marks distribution.
- Weak-concept weighting can personalize a paper for a specific student.
- Papers download as PDF; pricing is managed by admins under Exam paper pricing.
- Generated papers can be linked to courses and batches.

How to work in this module

1. Open Practice papers (students) or Paper generator (instructors/admins).
2. Choose the blueprint parameters and generate.
3. Review the paper and download the PDF.

2.7 Internships and community

The internship marketplace connects students with company listings. Community features - blogs, the wall of fame and public instructor profiles - keep learners engaged.

Key features

- Internship listings with filters; detail pages describe role, stipend and application flow.
- Students apply with their profile; companies and SPOCs manage requests from their portals.
- Blog with articles from instructors, students and SPOCs; newsletter subscription.
- Wall of fame highlights top learners; public instructor profiles list their courses.

How to work in this module

1. Open Internships and filter the listings.
2. Open a listing, review the requirements and apply.
3. Track responses under Internships and Internship Inbox in the student dashboard.

2.8 Cart, checkout and payments

Commerce flows through one cart and one checkout. Payments are processed by Razorpay; fulfillment is triple-redundant so a paid order is never lost.

Key features

- Cart holds courses, bundles, eBooks and exam papers; quantities and totals update live.

- Coupons apply at checkout where allowed (not on bundles or seat-priced cohorts).
- Razorpay checkout with signature verification; browser verify, webhook and a reconciliation sweeper all converge on the same fulfillment step.
- Receipts and invoice history; refund requests follow the published refund policy.
- Abandoned-checkout reminders recover lost sales automatically.

How to work in this module

1. Add items to the cart from any product page.
2. Open the cart, apply a coupon if you have one, and proceed to checkout.
3. Complete the Razorpay payment; access is granted within moments.
4. If the browser closes before verification, the webhook and sweeper still fulfill the order.

Payment failures surface an honest error state - nothing is granted until the gateway signature verifies.

PART 3

Student Workspace

3.1 Student dashboard

The dashboard is the learner home base: it summarizes progress, next actions and quick access to every learning surface.

Key features

- Continue learning strip that resumes your most recent lesson.
- Stat cards: active courses, completions, streak and XP.
- Shortcuts to learning plan, live classes, grades and mastery.
- Announcements and review reminders from your instructors.

How to work in this module

1. Sign in and land on the dashboard.
2. Use Continue learning to jump straight back into your last lesson.
3. Scan the stat cards; open any shortcut to continue working.

3.2 My courses and the lesson player

My Courses lists everything you are enrolled in. The lesson player is a unified reader for every lesson type with progress tracking and behavior signals.

Key features

- Lesson types: video (YouTube, streamed, Bunny CDN, audio-only toggle), text, H5P interactions, learning games, virtual labs, 3D models and 3D tasks, and recording lessons.

- Progress is marked per lesson; completing lessons advances course progress and can award XP.
- In-lesson tools: notes, playback speed, rewind and replay tracking.
- Adaptive practice and struggle heatmap panels appear beside non-quiz lessons.
- Offline learning bridge caches lessons for low-connectivity use.

How to work in this module

1. Open My courses and select a course.
2. Work through the curriculum in order; locked lessons unlock as prerequisites complete.
3. Inside a lesson, use the player controls; your interactions feed your mastery profile.
4. Mark the lesson complete when finished to update course progress.

Signals (rewinds, replays, pauses, early quits) are collected privately and only ever shown to you and your instructors as aggregate learning insight.

3.3 Quizzes and adaptive practice

Quizzes assess understanding with server-side timing, integrity checks and per-question feedback rules. Adaptive practice builds a personalized set from your weakest concepts.

Key features

- Question types: single choice, multiple choice, true or false, short answer and more; randomized option order where configured.
- Server-side timer with deadline enforcement; attempts and time windows come from the instructor settings.
- Integrity signals (tab switches, rapid answers) are recorded and shown to instructors as advisory data.
- After submission: score, correct answers (when the reveal policy allows), and weak concepts derived from wrong answers.
- Adaptive practice: choose focus concepts or let the engine pick; every session grades server-side.

How to work in this module

1. Open a quiz from the course curriculum and press Start.
2. Answer each question; the timer runs server-side, so do not leave the tab until you submit.
3. Review the result screen; read the explanations for wrong answers.
4. Open Adaptive practice (on My Mastery or in a lesson) and run a focused set on your weak concepts.

Answer keys never leave the server before submission - the pre-submit payload contains no correct answers.

3.4 Assignments

Assignments are teacher-graded work: file submissions, free-text answers and rubric scores.

Key features

- Assignment detail shows the brief, due date, maximum score and allowed attachments.
- Submit files or text before the deadline; resubmission follows the instructor policy.
- Graded results appear with rubric feedback under My Grades.
- Late submissions are flagged to the grader.

How to work in this module

1. Open the assignment from the course page or dashboard.
2. Attach your files or write your answer and press Submit.
3. Watch My Grades for the graded result and feedback.

3.5 My grades and the cumulative grade

My Grades consolidates quiz scores, assignment scores, interactive modules and live-class participation into one cumulative result per course.

Key features

- Cumulative grade weights: quizzes 40 percent, assignments 40 percent, interactive modules (games, H5P, labs, 3D tasks) 20 percent.
- Practice-only items are excluded from grading but still earn XP.
- Per-item breakdown with attempt history; gradebook CSV is available to instructors.
- Published exam results from your institution join the same page for campus students.

How to work in this module

1. Open My Grades from the Progress group.
2. Select a course to see the item-by-item breakdown.
3. Compare scores across categories to decide what to practice next.

3.6 My Mastery and learning signals

The mastery graph tracks your estimated confidence per concept from every scoring event - quizzes, assignments, labs, games, 3D tasks and live classes - each weighted by recency and source reliability.

Key features

- Concept cards show estimate (0-100), confidence and the evidence trail behind each number.
- Struggle profile explains, in human language, where and why you struggled.
- Lesson heatmaps visualize rewind and replay hotspots per lesson.
- Weak-concept lists drive adaptive practice, planner checks and AI tutor context.
- Prerequisites map shows how concepts build on each other.

How to work in this module

1. Open My Mastery from the Progress group.
2. Sort by weakest concepts; open one to see its evidence list.
3. Use the heatmap on a concept to find the exact lessons to review.

Mastery is derived, never hand-entered: every estimate can be traced to the exact quiz answers or lab runs that produced it.

3.7 My learning plan

The personal planner turns goals into scheduled study blocks with automatic review and delayed re-checks, replacing cramming with spaced practice.

Key features

- One active goal per course (for example, an exam date or weekly hours).
- Plan blocks for curriculum study, review of weak concepts and practice sets.
- Mark studied confirms work; cancelled reviews restore unfinished curriculum time.
- Failed or unclear checks go to your instructor intervention queue, not silently dropped.

How to work in this module

1. Open My Learning Plan and create or update a goal for a course.
2. Work the scheduled blocks; press Mark studied when a block is done.
3. Complete scheduled practice checks - success schedules a three-day delayed check to prove retention.

3.8 Live classes and recordings

Live classes are scheduled video meetings on a self-hosted Jitsi stack, joined with room-pinned tokens. Attendance, polls and (with policy) recordings are captured automatically.

Key features

- Upcoming classes with join buttons that open exactly during the scheduled window.
- In-class polls with live results; attendance is recorded from join and leave events.
- Recordings (when the class policy allows) appear with transcripts and AI topic summaries.
- Recording retention is policy-bound: media expires on schedule, knowledge (reports) stays.
- Ended classes link to recording notes and the class report.

How to work in this module

1. Open Live classes and join an upcoming class at start time.
2. Participate in polls; your attendance is tracked while you stay in the room.
3. After the class, open the recording reader for notes, transcript and playback.

Students never receive moderator or recording rights in class tokens.

3.9 Gamification: XP, badges, streaks and leaderboard

An event-sourced XP ledger rewards real completions across every feature. Levels, badges, streaks and an opt-out leaderboard make progress visible.

Key features

- XP for lesson completions, quiz results, games, labs, 3D tasks, live attendance and teach-backs.
- Levels follow the formula $100 \times n \times (n + 1) / 2$ XP per level n .
- Around fifteen seeded badges plus custom per-course badges; freezes protect streaks on missed days.
- Leaderboard is opt-out and never exposes emails; course scopes can switch it off.

How to work in this module

1. Earn XP automatically as you complete work - there is nothing to claim.
2. Check the leaderboard (Progress group) and opt out there if you prefer.
3. Keep a daily streak alive; streak freezes are consumed automatically on a missed day.

3.10 Certificates

Certificates are issued on course completion from instructor-designed templates, with a public verification page and QR code.

Key features

- Automatic issuance rules per course; PDF download from your profile.
- Public verify page (by certificate ID) shows issuance details without exposing personal data.
- Revocation by admins for refunded or revoked completions.

How to work in this module

1. Complete a certificate-bearing course.
2. Download the certificate from the course or profile page.
3. Share the verification link - employers can confirm authenticity themselves.

3.11 Career center: internships, vouchers and messages

The career group manages internship applications, work documents and messaging.

Key features

- My internships: active applications, accepted roles and work logs.
- Internship inbox for company and SPOC correspondence.
- Vouchers issued by companies or SPOCs under My vouchers.
- Messages: direct conversations tied to your applications.
- Student blog: write and publish articles.

How to work in this module

1. Apply to internships from the public listings.
2. Track state changes under My internships; respond in the inbox.
3. Log daily work once accepted; managers review the logs.

PART 4

Instructor Workspace

4.1 Instructor dashboard

The teaching overview aggregates courses, learners, grading load and reviews in one screen.

Key features

- Stat tiles: active courses, total students, purchases, pending reviews, average rating.
- Pending grading queue with direct links.
- My courses grid with per-course enrollment counts.
- Today review queue: learner questions, error reports and AI drafts.
- Export buttons for courses, students, quiz results and assignment results (CSV).

How to work in this module

1. Sign in as an instructor and review the tiles.
2. Open the grading queue when the pending count is above zero.
3. Use Open your campus when you also manage an institution workspace.

4.2 Creating a course

Course creation is draft-first: a short start form collects the essentials, then the full tabbed editor opens. Course type decides which tools are available.

Key features

- Start form: title, description, type, pricing (free, one-time or sale price).
- Course types (Meiporul, Seyappaduporul, Utporul) map to a capability matrix of allowed tools.
- Publishing is explicit; drafts are invisible to learners.
- Type capabilities are exposed publicly so the wizard can mirror them.

How to work in this module

1. Dashboard, Create New Course (or My courses, create).
2. Fill the start form and save; the draft opens in the editor.
3. Set pricing and details in Basic info.
4. Build curriculum, then publish from settings when ready.

Changing price later does not affect past buyers; sale display rules follow a single server-side source of truth.

4.3 The course editor

One tabbed editor manages the whole course: content, collaborators, scheduling, studio faces and publication.

Key features

- Tabs: Basic info (with tags editor and custom link), Curriculum, Collaborators, Settings, Studio.
- Deep links with ?tab= and ?face= open a specific tab or studio face directly.
- Co-instructors and collaborators can be granted edit access (the platform has one shared edit rule).
- Studio faces: parent view and reward settings per course, with dismissible onboarding.
- Preview-as-learner shows exactly what students see, including locks.

How to work in this module

1. Open My courses, then edit a course.
2. Work through the tabs; every save is immediate.
3. Add collaborators by email with an explicit role.
4. Use Publish in Settings when the curriculum is complete.

4.4 Lesson types and authoring

Lessons are typed content units. The editor enforces which types each course type may use and validates every attachment as an atomic pair (type plus reference).

Key features

- Video lessons: YouTube link, server-streamed unlisted video, or Bunny CDN upload; audio-only toggle available.
- Text lessons with rich content and concept links.
- Interactive: H5P package, learning game, virtual lab, 3D model and 3D task.
- Recording lessons: converted from live-class recordings with transcript review.
- Any lesson can be marked as a free preview for visitors.

How to work in this module

1. In the editor, open Curriculum and add a lesson.
2. Choose the lesson type; the form adapts to it.
3. Attach the required content item (video URL, game, lab, model or package).
4. Mark preview lessons deliberately - they are public.
5. Save, then reorder the curriculum as needed.

Publishing a lesson type without its attached item is rejected - the pair is validated together in one request.

4.5 Quiz builder and question banks

The quiz builder creates assessments with questions, interactive modules and integrity settings. Question banks store reusable, tagged questions with import and item analysis.

Key features

- Question editor with per-question points, explanation and concept links; CSV import for bulk authoring.
- Interactive modules: attach H5P, games, labs or 3D tasks as scorable items with weights.
- Reveal policies (never, after submission, after deadline) protect answer keys.
- Random per-attempt subsets and option shuffling; retired questions drop out of new attempts only.
- Banks: snapshot copy from quizzes, AI-drafted questions land tagged as drafts and never auto-publish.
- Item analysis reports facility and discrimination per question.

How to work in this module

1. Open Quiz builder from a course or the assessment group.
2. Create the quiz shell: timing, attempts, reveal policy, pass mark.
3. Add questions manually or import from a bank or CSV.
4. Attach interactive modules as graded or practice-only items.
5. Publish the quiz; it now appears in the course curriculum and gradebook.

A quiz without questions is valid when it carries interactive modules; maximum scores for modules are always derived server-side.

4.6 Assessment Studio

Assessment Studio produces exposure-aware practice sets from bank content with immutable snapshots, so published practice never changes under learners.

Key features

- Studio questions reference bank items but freeze a snapshot at publication.
- Version history, review signatures and retirement without touching existing sessions.
- Exposure-aware selection prefers items the learner has seen least.
- Adaptive negative IDs map to snapshots; positive IDs remain ordinary quiz questions.

How to work in this module

1. Open Assessment Studio (deep-linked from interventions or a course).
2. Choose course and concept; compose the set from the bank.
3. Submit for review, then publish; scheduled sets reach learners through the planner.

4.7 Grading, gradebook and interventions

Manual grading covers assignments and free-text quiz answers with rubrics. The gradebook aggregates every scorable item; interventions act on learner struggle.

Key features

- Grading queue with rubric scoring and inline feedback; grading lock prevents double-grading.
- Gradebook: cumulative course grades, per-student breakdown, CSV export.
- Insights hotspots: aggregated struggle heatmaps per course and segment alerts.
- Interventions: failed planner checks and at-risk flags arrive as an actionable queue.
- Review queue: learner questions, content error reports and AI drafts waiting for you.

How to work in this module

1. Open Grading and pick the oldest submission.
2. Score against the rubric and write feedback; submit to release the result to the student.
3. Weekly, open Insights to check hotspots and segment alerts.
4. Work the Interventions queue: assign practice, message the learner or resolve the check.

4.8 Games builder

Instructors build learning games from five templates without any files: quiz rush, match pairs, drag and sort, word builder and sequence.

Key features

- Strict per-template validation with live blocking checks and duplicate warnings.
- Maximum score is always derived (10 x item count) - never authored.
- Games publish to a marketplace; admin prebuilt packs also appear there.
- Attach games to lessons or quizzes; results feed XP and mastery as engagement data.

How to work in this module

1. Open Games in Content Studio and create a game.
2. Pick a template and fill its config; validation guides you field by field.
3. Preview, then publish; attach it to a lesson or quiz.
4. Review results in analytics (advisory, not gradebook truth, unless attached as a graded module).

4.9 H5P library

H5P packages authored in external tools (such as Lumi) are uploaded once and reused across lessons and quizzes.

Key features

- Upload .h5p archives with server-side safety checks (traversal, symlink, zip bomb, extension guards, per-owner cap).

- Player runs package scripts inside a sandboxed iframe - never with same-origin rights.
- Client-reported scores are advisory engagement data; graded use goes through scorable items.

How to work in this module

1. Author the package externally and export it.
2. Open H5P Library, upload and let validation finish.
3. Attach the package to a lesson or quiz module.

4.10 Lab Studio

Lab Studio authors guided labs and manages the lab catalog for your courses: built-in simulations, native labs and supplied packs.

Key features

- Guided lab authoring: steps, checkpoints and questions around a simulation.
- Native engines (reaction lab, identify lab) with server-validated answer keys.
- Import supplied packs (for example CBSE concept extensions) and curriculum mapping.
- Labs attach to lessons as a type-plus-slug pair; unpublished slugs are rejected.

How to work in this module

1. Open Lab Studio in Content Studio.
2. Create a guided lab around a chosen simulation, or import a pack.
3. Publish, then attach to a course lesson.

4.11 3D teaching kit and 3D tasks

The 3D kit uploads GLB models with tier processing (automatic optimization), and 3D tasks turn models into graded match-and-verify exercises.

Key features

- GLB upload with magic-byte validation, size caps and per-tier budget checks.
- Tier files (optimized variants) are generated by the media pipeline and listed per model.
- Seven task types: match, identify, verify, assemble, measure, manipulate, sequence.
- Anchors are normalized region boxes; labels never leak to learners.
- Grading is server-side from submitted answers; evidence events produce a confidence signal.

How to work in this module

1. Upload a GLB under 3D models and wait for tier processing.
2. Open 3D Tasks, create a task and choose its type.
3. Define anchors on the model; configure scoring.
4. Publish and attach to a lesson or quiz; learners earn the same marks in list view without WebGL.

4.12 Recording lessons

Recording lessons turn live-class recordings into curriculum: transcribe, review, then convert into a draft lesson.

Key features

- Private workbench lists recordings from your live classes.
- Transcription worker (optional isolated environment) produces transcripts.
- Review pass: correct segments, tag concepts, choose publication.
- Conversion creates one normal draft lesson with source metadata; questions become Assessment Studio drafts.

How to work in this module

1. Open Recording Lessons in Content Studio.
2. Select a recording and generate or wait for the transcript.
3. Review and correct, then convert to a lesson draft.
4. Finish and publish the lesson from the course editor.

4.13 Live class console

The console schedules and runs live classes: Jitsi video, polls, attendance and post-class reports with retention controls.

Key features

- Schedule single or recurring classes with purpose, mode, audience and recording policy.
- In-class: polls with live results, attendance tracking and event log.
- End of class produces a permanent report (engagement, polls, notes) independent of recording retention.
- Recording lifecycle: soft delete with 30-day restore, retention extension under the ceiling, full audit.

How to work in this module

1. Open Live classes, schedule a class with a room and time.
2. Start the class at the scheduled time; learners join from their side.
3. Run polls during the session.
4. End the class to generate the report; manage the recording from Past classes.

4.14 Students, analytics and insights

Learner-facing analytics for instructors: who is struggling, where, and what to do about it.

Key features

- Students list per course with progress and last-active columns.
- Course analytics: completion, scores, participation trends.
- Hotspots (signals) aggregate rewind and replay behavior per lesson segment.

- Coverage report per course: taught versus assessed concepts and outcome mapping.
- Cohort mastery panels for SPOC-style cohort views.

How to work in this module

1. Open Students or Analytics from Learners and Insights.
2. Sort by progress to find inactive learners.
3. Open Insights, Hotspots tab; deep-link to a struggling segment.
4. Use coverage to find concepts with no assessment, then fix the gap in the quiz builder.

4.15 Certificate designer

The designer creates certificate templates from elements (text, borders, QR, signatures) that render identically at issuance time.

Key features

- Visual canvas editor producing an elements config - the single source of truth for template and issuance.
- Template gallery with five seeded starters; thumbnails regenerate on Chrome-capable hosts.
- Issuance renders through the headless-Chrome pipeline with a ReportLab fallback.
- QR verification codes printed automatically.

How to work in this module

1. Open Certificates in Learners and Insights.
2. Start from a template and adjust elements.
3. Save and preview the render.
4. Configure the course to issue this template on completion.

4.16 Course upload, backup and restore

Course packages export a course (content only - never learner or payment data) to a portable file that can be restored as a new draft.

Key features

- Export produces a portable package with curriculum, assets and metadata.
- Upload and inspect packages privately; staged uploads stay outside public storage.
- Restore-as-new-draft recreates the course for editing; transfer history tracks ownership.
- Preview a package privately before restoring.

How to work in this module

1. Open Course Upload and Backup.
2. Export the course you want to back up.
3. To restore, upload the package, inspect it, then restore as a new draft.
4. Clean up expired staging from the same screen.

4.17 Ebooks and instructor blog

Ebooks authoring publishes titles to the digital library; the instructor blog publishes articles to the community blog.

Key features

- Ebook uploads with cover, sample and price; appear in the public library.
- Blog editor with categories, drafts and publishing workflow.
- Blog categories managed by admins; export to SPOC blog possible.

How to work in this module

1. Open My Ebooks, create or upload a title, set price and publish.
2. Open Blog, write an article, choose a category and publish.

PART 5

Campus OS for Institutions

The institution workspace is the heart of Campus OS. Owners and admins run the school; teachers see their batches; student members see their own campus view. All modules below live under /institutions/{id}/... and share one audit log, one member list and one action center.

5.1 Institution workspaces and people

Every school or college is a workspace with members (owner, admin, teacher, student), batches, connected courses and an audit trail.

Key features

- Create an institution with kind, academic year, timezone and plan.
- Invite members by email with a workspace role; deactivate or change roles later.
- All sensitive actions are recorded in the institution audit log.
- Role-filtered tabs: managers see admissions, finance and trust; staff see staff tools; learners see their own views.

How to work in this module

1. As owner, open Institutions and create or select the workspace.
2. Invite members from the People tab with the right role.
3. Configure batches and connect courses under Batches and Courses.
4. Check the audit log for a record of every configuration change.

5.2 Today and the Action Center

Today is the daily command center: role-aware priorities, the day schedule, notices and assignable action items in one view.

Key features

- Morning brief: schedule, attendance and grading reminders, fee alerts, admissions signals.
- System action cards: unverified cash, uncovered substitutions, routes without a boarding log, pending hostel passes.
- Covering teachers see substituted classes in their own schedule.
- Items are derived - completing the underlying task clears the card.

How to work in this module

1. Open the workspace and select Today.
2. Work top to bottom: schedule first, then system actions.
3. Click a card to jump to the module that owns the task.

5.3 Admissions

Admissions manages the funnel from enquiry to enrolled student: programs, intakes, applicants, documents, offers and conversion.

Key features

- Programs with intakes and capacity; applicant stages with full history.
- Document tracking per applicant, notes and tasks for the team.
- Offers with PDF output; enrollment conversion creates the student record.
- Pagination, optimistic concurrency (edit conflicts are detected) and idempotent writes.

How to work in this module

1. Define programs and intakes with capacity.
2. Capture applicants and move them through stages.
3. Track requested documents until complete.
4. Issue an offer (PDF); on acceptance, convert to an enrolled student.

5.4 Academics: batches, timetable and resources

Academics organizes batches, the shared timetable and learning resources for the campus.

Key features

- Academic batches per department and year; batch members.
- Timetable slots (campus events) with room and batch conflict checks.
- Substitutions appear on slots automatically when the owner is on approved leave.
- Resources tab shares campus learning materials.

How to work in this module

1. Create batches under Batches; add students to batches.
2. Schedule slots under Academics; conflicts are refused at save time.
3. Review the daily view under Daily campus.

5.5 Exams

Exams run the term examination cycle: papers on the timetable, hall tickets, marks entry and published results.

Key features

- Exams per academic term with draft, scheduled and published states.
- One paper per batch and subject, placed on the shared timetable with room and batch conflict checks.
- Hall tickets issued lazily and race-safe, with roll numbers and verification codes; branded PDF.
- Marks entry with absent handling, locked once results are published.
- Consolidated batch results with rank and pass or fail; branded marksheets; results join the term report card.

How to work in this module

1. Create the exam and its papers; schedule them onto the timetable.
2. Publish the schedule - hall tickets become visible to students and approved guardians.
3. On exam day, staff enter marks (absences flagged).
4. Publish results; marks lock, results appear to learners and guardians, and flow into report cards.

Learners and guardians see hall tickets once scheduled but results only after a manager publishes.

5.6 Staff, leave and substitution

Staff manages teacher leave with derived balances and automatic substitution planning.

Key features

- Leave types with annual quotas per academic year; balances are derived from approved days.
- Teachers apply for their own leave (no overlaps, at most 60 days).
- Managers approve against balance - an audited override note is required to exceed a quota - or reject with a reason.
- Approval opens substitutions for every timetable slot the absent teacher owns; cancel releases them.
- Candidate checks prevent double-booking: no overlapping class, substitution or leave.
- Leave and substitution-hours report with CSV export.

How to work in this module

1. Teacher: apply under Staff with dates and type.
2. Manager: review the balance and approve or reject.
3. Assign substitutes from the suggested free candidates; Covered by appears on the timetable.

4. Managers act on the Today card while classes in the next seven days are uncovered.

5.7 Transport

Transport manages routes, stops, student assignments and the daily boarding log.

Key features

- Routes with vehicle, driver, capacity and ordered stops with pickup and drop times.
- One active route and stop per student, capacity enforced.
- Daily boarding and drop log for today or past days; route roster CSV.
- Students and approved guardians see route, stop and today status.
- A route with a fee owns a published tuition plan (Transport, route name); assigning a student assigns that plan.

How to work in this module

1. Create the route and its stops.
2. Assign students to route and stop.
3. Record boarding daily (managers get a Today alert while an active route has no log).
4. Export the roster or review past days as needed.

5.8 Hostel

Hostel manages blocks, rooms, allocations, out-passes and visitors.

Key features

- Blocks with warden and optional fee; rooms with type and bed capacity; allocations with check-in and check-out, capacity enforced.
- Out-passes requested by residents; staff approve, reject or mark returned.
- Visitor register; occupancy report with CSV.
- Residents and approved guardians see room and pass state; staff get a Today action while passes are pending.
- Block fees follow the same plan-per-block rule as transport.

How to work in this module

1. Define blocks and rooms.
2. Allocate residents with check-in dates.
3. Residents request out-passes; staff process them from the Today card.
4. Log visitors; run the occupancy report monthly.

5.9 Finance: fee plans, assignments and the ledger

Student finance starts with fee plans: components and installments that must balance exactly, assigned to students with an append-only ledger.

Key features

- Fee plans: components (code, name, amount) and installment templates; totals must match to the paisa.
- Assignment to a student member creates the installment schedule.
- Discounts and waivers as adjustments; every posting is an append-only ledger row.
- Balances and aging per student; parent and student views; role-scoped access.
- Transport and hostel fees integrate through their own plan-per-item rule.

How to work in this module

1. Build the plan: components first, then installments that sum to the same total.
2. Assign the plan to students (individually or by batch).
3. Apply discounts or waivers with a reason when approved.
4. Review balances and aging under Finance; the ledger never rewrites history.

5.10 Finance: fee collection, receipts and invoices

Fee collection covers money at the counter and online, with verification and branded paperwork.

Key features

- Cash and cheque recorded at the counter carry the receiving staff member and stay pending until a second owner or admin verifies the count.
- The receiver cannot verify their own entry; a one-person office may self-verify with an audited tag.
- Daily cash desk per receiver with CSV export; a Today action fires while cash is unverified.
- Branded receipt PDFs for every payment showing receiver and verification.
- Numbered demand invoices for one installment or the whole balance, printable by managers, students and approved guardians.
- Online payment through Razorpay from student and parent fee pages: signature-verified, reconciled, fulfilled through the same ledger path.
- Over-capture posts up to the balance and flags the remainder for manual refund.

How to work in this module

1. Counter: record the payment with method, amount and received-by.
2. A second manager verifies from the cash desk (or self-verify when alone, with the audit tag).
3. Print or download the receipt PDF.
4. Invoice: raise a demand invoice when needed; the student or guardian pays online.
5. Online: student or guardian presses Pay online, completes Razorpay, and the ledger updates automatically.

Until gateway keys are configured the online path returns an honest service-unavailable message; counter collection keeps working.

5.11 Fee reminders

One reminder policy per institution drives automatic fee reminders through email and WhatsApp with a full delivery ledger.

Key features

- Policy: days before due, overdue cadence and cap, send hour in campus time, channels and WhatsApp template.
- Recipients: the student plus approved guardians.
- WhatsApp only for consented campus members; everyone else falls back to email; unusable channels are recorded as skipped with a reason - never as sent.
- Append-only ledger with one row per installment, stage, recipient and channel - re-runs never send twice.
- Managers can run now, retry failed rows or send a manual reminder from an account row; receipts queue notices after every payment.

How to work in this module

1. Configure the policy once under Finance, Reminders.
2. The campus worker sends reminders automatically each tick.
3. Watch the delivery ledger; retry failures or send manual reminders as needed.

Prerequisites: SMTP for email; WhatsApp Cloud credentials plus an approved template saved in the policy for WhatsApp reminders.

5.12 Progress and reports

The Progress tab reports learning outcomes for the campus: batch progress, attendance and published exam results joining the term report card.

Key features

- Batch-level progress across connected courses.
- Attendance views for the last 30 days.
- Published exam rows feed the term report card automatically.
- Guardian-safe: parents see only their approved children.

How to work in this module

1. Open Progress and choose the batch.
2. Drill into a student for the consolidated view.
3. Use term report cards at parent meetings.

5.13 Plan, usage and settings

Managers administer the workspace subscription and its settings.

Key features

- Plan and usage: seats, storage and feature consumption against the institution plan.
- Settings: workspace identity, defaults and academic year.
- Plan changes go through the platform plan queue (admin approval).

How to work in this module

1. Open Plan and usage to review consumption.
2. Request a plan change; track it in the queue.
3. Adjust workspace settings as the year turns.

5.14 Trust and integrations

The trust center covers custom domains, integrations, privacy and data portability.

Key features

- Custom-domain flow with the exact DNS record to add.
- Integration configuration stores only secret references - plaintext provider secrets never live in the database.
- Privacy requests and consent receipts; retention rules per data class.
- OneRoster 1.2 ZIP export for roster interchange.

How to work in this module

1. Open Trust and integrations as a manager.
2. Add the DNS record shown, then verify.
3. Enter integration reference names (from your secret manager) and enable.
4. Handle privacy requests and run OneRoster exports from the same screen.

5.15 Campus growth and the public campus page

Growth tools market the institution: a public page, plan comparison, consented lead capture and a help center.

Key features

- Public /campus page with plan comparison and lead form (consent recorded).
- Lead queue inside the growth control plane for follow-up.
- Searchable help center for product questions.
- Branded share banners for campaigns.

How to work in this module

1. Publish the institution profile to the public page.
2. Collect leads; follow up from the queue.
3. Point families at the help center for self-service.

5.16 WhatsApp communications

The communications center runs WhatsApp campaigns and transactional sends platform-wide, with compliance built in.

Key features

- Global center at /admin/communications; institution workflows reuse the same pipeline.
- Templates (approved Meta templates), delivery state, retry and rate controls.
- Opt-in records, quiet hours and full audit trails.
- Campus notifications (fee reminders, notices) send through one-recipient campaigns.

How to work in this module

1. Admin sets up Meta WhatsApp Cloud credentials and approved templates.
2. Create the campaign, choose the audience, schedule inside quiet-hour rules.
3. Track delivery per recipient; retry failures from the ledger.

Members without WhatsApp consent are never messaged on WhatsApp - the system falls back to email and records the choice.

PART 6

Company Portal

6.1 Company dashboard and people

Companies get their own workspace for internship programs, with owners and managers scoped strictly to their own company data.

Key features

- Owner creates the company profile; approval by a platform admin unlocks the portal.
- Managers invited by the owner; access is company-scoped (managers never see other companies).
- Announcements to enrolled interns.
- Seat pools from paid invoices; managers assign seats to users.

How to work in this module

1. Register the company (or For companies, sign up) and await approval.
2. Owner invites managers under People.

3. Publish announcements to your interns.

6.2 Internship management

The operational core for company internships: listings, requests, work logs, attendance and performance reviews.

Key features

- Create internship listings; receive and process applications.
- Work logs: daily entries by interns, reviewed by managers.
- Attendance tracking and performance reviews per intern.
- Vouchers and company notes; bulk invite students into a cohort.

How to work in this module

1. Create a listing with role details and stipend.
2. Review requests and accept candidates.
3. Track daily work logs and attendance; run periodic reviews.
4. Issue vouchers or complete internship records on finish.

6.3 Company invoicing and seats

Admin-negotiated GST invoicing pays for company seat pools.

Key features

- Admin raises a numbered GST invoice (settlement is idempotent across verify, webhook and manual mark-paid).
- Paid invoices create a company seat pool.
- Managers assign seats to existing users; enrollment is granted with a company source that memberships can never suspend.
- Invoice PDFs are private and authenticated.

How to work in this module

1. Admin creates the invoice under Admin, Invoices.
2. Company pays online or is marked paid offline.
3. Managers assign the seats from the dashboard.

PART 7

SPOC Workspace

7.1 SPOC dashboard and cohorts

The Single Point of Contact runs placement cohorts for a college: a dashboard, cohort management, internship coordination and a blog.

Key features

- Dashboard with cohort statistics and placement progress.
- Cohorts per college and course; seat pricing supported for referral checkout.
- Internship management mirrored from the company side for coordination.
- SPOC blog with export.

How to work in this module

1. Sign in as SPOC and open the dashboard.
2. Create or select the cohort; enroll students.
3. Coordinate internships between companies and students.
4. Publish cohort news on the blog.

PART 8

Parent Portal

8.1 My children

The parent portal aggregates every approved child into one page: attendance, fees, results, transport, hostel and notices - plus the online-learning digest.

Key features

- Approval-based linking: a parent sees a child only after the link request is approved.
- Fees: outstanding and overdue amounts, next due installment, Pay now, invoices and receipts.
- Attendance for the last 30 days.
- Latest published exam results with rank; transport status for today; hostel room and out-pass state.
- Derived alerts: overdue fees, low attendance, not boarded, pending out-pass, newly published results.

How to work in this module

1. Sign in as a parent; request a link to your child (approved by the institution).
2. Review each child card top to bottom.
3. Press Pay now to settle fees online; download invoices and receipts.
4. Act on alerts - each links to the underlying detail.

Every block degrades independently: if one campus service is unavailable, its card shows nothing rather than breaking the page.

PART 9

Platform Administration

9.1 Admin dashboard and Operations Center

The admin dashboard is the platform cockpit; the Operations Center consolidates daily operations, outcomes, revenue and system health into one page.

Key features

- Daily operations queues: approvals, refunds, support.
- People and courses oversight with consistent lists and CSV exports.
- Learning outcomes reporting (verified outcomes, not vanity metrics).
- Revenue: cash, refunds and subscription reporting.
- System health: payment health pipeline and background worker status.

How to work in this module

1. Open Operations Center from the admin sidebar.
2. Work the queues top to bottom.
3. Check payment health after gateway incidents; the reconciliation sweeper self-heals missed webhooks.
4. Export CSVs for finance and audit.

9.2 People and access

User administration for the whole platform.

Key features

- Students and instructors directories with search and filters.
- Approvals: instructor approvals and company approvals.
- Manage: user administration including password-reset links, role changes and 2FA administration.
- Enrollments: search, status changes, bulk status updates and admin-granted enrollments.
- Membership cancellation and refund handling where applicable.

How to work in this module

1. Open People and Access, then the relevant directory.
2. Search for the user; open the row.
3. Apply the action (approve, reset link, change role, grant enrollment).

4. For bulk work, use the bulk status tool with CSV.

9.3 Commerce administration

Everything sold on the platform is administered here.

Key features

- Orders: full order history with per-course items and bundle breakdowns.
- Coupons: codes with windows, limits and Razorpay offer linkage for memberships.
- Memberships: tier CRUD (new price equals new tier), subscriber lists, cancellation.
- Bundles: CRUD with course membership; paid buyers keep their snapshot forever.
- Company invoices: create, settle, download.
- Exam paper pricing for the public paper store.

How to work in this module

1. Open the Commerce group and the relevant tool.
2. Create or edit the item; pricing rules are enforced (immutable gateway plans, no coupons on bundles).
3. Audit disputes from Orders using the payment-health view.

9.4 Learning and content administration

Platform-wide content governance.

Key features

- All courses: moderation, publish state, owner reassignment.
- Categories and tags (with the derived tag-cluster panel).
- Reviews moderation.
- Content Libraries: import virtual labs, prebuilt games and 3D models (all-or-nothing imports).
- Certificates: issuance records and revocation.
- Admin Lab Studio, Course Upload and Backup, and the paper generator mirror instructor tools platform-wide.

How to work in this module

1. Open Learning and Content.
2. Review flagged courses or reviews first.
3. Import library packs as needed - validation rejects nothing halfway.
4. Use tag clusters to keep search clean.

9.5 Communications, messages, blogs and wall of fame

Community and messaging surfaces for the platform.

Key features

- Communications Center: WhatsApp and email operations (see 5.16).
- Messages: platform message monitoring.
- Blogs: category management and moderation.
- Wall of Fame: curate showcased learners.

How to work in this module

1. Open the Insights and Community group.
2. Moderate content and messages.
3. Curate the wall of fame periodically.

9.6 Analytics and settings

Platform analytics and global settings.

Key features

- Analytics: enrollment, revenue and engagement dashboards.
- Settings: platform configuration, feature switches and defaults.
- Export and import (CSV transactional export/import).

How to work in this module

1. Review analytics weekly.
2. Change settings deliberately - most switches are platform-wide.

9.7 Superadmin

The supervisory role above admin: monitoring dashboards and impersonation audit, with access to the full admin console.

Key features

- Overview: platform-wide health and activity.
- Students, instructors and admins monitoring views.
- Impersonation audit: every admin view-as action is logged.
- Direct link into the admin console when action is required.

How to work in this module

1. Sign in as superadmin (TOTP required).
2. Review the four dashboards.
3. Audit impersonation events; investigate anomalies.

PART 10

Cross-Cutting Platform Services

10.1 Notifications and communication preferences

Every role receives in-app notifications and optional email; users control what they get.

Key features

- In-app notification center with unread counts on the sidebar.
- Retention rules (streaks, inactive learners, hot segments) send notifications, never surprise emails.
- Communication preferences page per account: category-level opt-in and opt-out.
- Guardian digests only include the signals the course studio ticked.

How to work in this module

1. Open Communication preferences from the sidebar.
2. Toggle categories to taste; save.
3. Clear the notification center as you work.

10.2 Search

Search reaches courses, tags (with cluster expansion), labs and the library.

Key features

- Header search on the public site and in dashboards.
- Course search matches title, body and tags - including every member of the matched tag cluster.
- Tag suggestions for instructors while authoring.

How to work in this module

1. Type in the search box and submit.
2. Results rank by relevance; cluster members keep related courses discoverable.

10.3 Exports and data portability

Data leaves the platform through governed exports.

Key features

- CSV exports: gradebook, enrollments, orders, operations lists, leave reports, cash desk, occupancy.
- Course packages: portable, content-only course backups.
- OneRoster 1.2 ZIP: standards-compliant roster interchange.
- Instructor self-exports from the dashboard (courses, students, quiz and assignment results).

How to work in this module

1. Find the export button in the owning module.
2. Exports are permission-checked and never include other tenants data.

10.4 Mobile and offline learning

A Flutter application mirrors the learning experience, with an offline bridge on the web.

Key features

- Flutter app: dashboard, courses, live classes, quizzes, planner and recording reader.
- Offline learning bridge caches lessons; progress syncs on reconnect.
- Offline page handles connectivity loss gracefully.

How to work in this module

1. Use the mobile app with the same account.
2. Download lessons for offline study; the bridge syncs automatically.

10.5 Accessibility and localization

The platform is bilingual (English with Tamil taxonomy) and ships an accessibility helper.

Key features

- Accessibility assistant widget on every page (contrast, text size).
- Tamil category taxonomy (Meiporul, Seyappaduporul, Utporul) is intentional product naming, not untranslated text.
- Fonts include Tamil (Noto Sans Tamil) for native rendering.

How to work in this module

1. Open the Accessibility button on any page.
2. Adjust settings; they persist for the session.

APPENDIX

Reference

A. Quick route reference

Area	Route
Sign in / register	/login, /register
Student dashboard	/dashboard
My courses / plan / mastery / grades	/my-courses, /my-plan, /my-mastery, /my-grades

Area	Route
Live classes (student)	/student/live-classes
Instructor dashboard / courses	/instructor/dashboard, /instructor/courses
Course editor	/instructor/courses/{id}/edit
Live classes (instructor)	/instructor/live-classes
Grading / insights / interventions	/instructor/grading, /instructor/insights, /instructor/interventions
Institution workspace	/institutions/{id} (tabs: today, admissions, finance, exams, staff, transport, hostel, trust)
Parent portal	/parent
Company portal	/company/dashboard
SPOC workspace	/spoc/dashboard
Admin console	/admin/dashboard (operations, people, commerce, content, communications)
Superadmin	/superadmin/dashboard
Public campus page / help	/campus, /help

Table 2. Frequently used routes by area.

B. Running the platform locally

Step	Command
Start backend (SQLite preview, port 8000)	backend/.venv/Scripts/python.exe scripts/preview_campus.py
Start frontend (Vite dev server, port 3000)	cd frontend and run: npm run dev
Preview accounts	.local/campus-role-logins.json (10 role accounts)
Health check	http://127.0.0.1:8000/health
API documentation	http://127.0.0.1:8000/docs (development mode)

Table 3. Local development runbook (from the repository root).

The local preview uses a synthetic campus (Greenwood College) with seeded students, batches, courses and every account role. It never reads production environment files and keeps its database and credentials under the ignored .local directory.

C. Glossary

Term	Meaning
XP event	Append-only ledger row keyed for idempotency; the single source for points, badges and streaks.
Mastery estimate	Recency-weighted confidence score (0-100) per concept, derived from scoring evidence.
Scorable item	A graded attachment (H5P, game, lab, 3D task) with a server-derived maximum score and weight.
Enrollment source	How access was granted: purchase, bundle, membership, company seat or admin grant; suspends only where allowed.
Snapshot	Immutable copy of bank content used by published practice so later edits never change live sets.
Campus event	A timetable slot (class, exam) owned by a teacher or batch, used for conflicts and substitutions.
Ledger row	Append-only finance posting; corrections are new rows, never edits.
Retention (recordings)	Policy-bound lifetime of class media; reports and knowledge never expire.
Reconciliation sweeper	Background pass that heals missed webhooks by comparing with the payment gateway.
Studio face	Per-course parent-view and rewards configuration shown in the course editor.

Table 4. Platform vocabulary.